

ObsSec Star Hopping Guide for Michaelmas 2026

There are two main reasons why this guide was created:

- Firstly, star-hopping is really, really fun! I had never done much visual astronomy before joining CUAS, and getting my Northumberland license was pretty cool... then I learnt how to star hop, i.e. how to use a star map to get from A to B across the sky. Along the way I've seen clusters, globular clusters, planets, stars, galaxies, nebulae and so much more! I hope that learning how to star hop will improve your familiarity with the night sky, and show you just how cool navigating without right ascension and declination coordinates can really be.
- Secondly, I wanted this guide to help bridge the gap between using the Northumberland and using the Thorowgood. The Northumberland's nice and friendly: there's a computer that gives you coordinates for whatever you want to look at. The Thorowgood is a different story. There's right ascension dials that no one knows how to use... so your only real option if you want to find anything is to star hop. But then you only have a narrow slit to look through, the telescope itself has lots of rods, you have to do a meridian flip, and all of a sudden the task seems a little intimidating.

Let me introduce you to the Dobsonian: the ground reflector telescope that you probably didn't know existed in the Northumberland cupboard. It's a lot easier to set up, move around, and you can still see plenty of cool things through it. So, what better place to start building your star-hopping skills than with the Dobsonian.

This guide contains 6 paths through the sky that you should be able to follow using star hopping. There are 2 easy, 2 medium and 2 hard paths. All paths start off by pointing the Dobsonian at an easy target, usually a bright star in a constellation. You shouldn't need any fancy star maps or Sky Safari Pro. Instead, all of these paths should be followable using Stellarium (spoiler, Stellarium is definitely not the best star map app out there, but it's free and perfectly good enough for our purposes. Besides, if you can star hop using stellarium, other star maps will be so much easier to use).

Also, expect to find star hopping a little tricky at first - getting used to the way the Dobsonian works, what stars actually look like through the telescope compared to Stellarium, getting angry at Stellarium, getting lost and having to start over - all of these are part of the process. But the reward is great - you'll know so much more about what's where in the night sky, all while developing what's in my opinion the coolest skill in visual astronomy.

Any comments or suggestions are very welcome - send them to the ObsSec email and we'll use them to improve later editions of this guide.

Clear skies and best of luck!

Lan ObsSec, CUAS

(The routes in this edition of the Star Hopping Guide were created by Lan Hall and William Moorhouse, ObsSecs 2026-27.)

Using the Dobsonian

Firstly, to access the Dobsonian, you'll need to use the key attached to the Northumberland key card to open the cupboard door in the Northumberland (so, I guess step 0 is to get a Northumberland license).

More information about the Dobsonian is also included in the telescopes guide on the CUAS website, but here's a quick round up of what you need to know:

Setting up and taking care of the telescope

- The Dobsonian comes in two parts, the base and the telescope tube. When getting out of the cupboard and onto the lawn, it's easiest to take the telescope tube off the base and carefully place it upright next to the base. Then move the base outside, and finally put the telescope tube on top. Set it up on the lawn and not the path so you don't get run over by a cyclist.
- The Dobsonian only balances itself when the lens cap is off, so take the cap off before leaving the telescope at an angle, otherwise it just tips over. When changing eyepieces and things, you can use the cap as a tray to keep eye-pieces in, but be careful not to damage anything!
- The eyepieces are in a wooden box in the same area as the Northumberland eyepieces - use the Meade eyepieces. The eyepiece holder doesn't hold very securely, so make sure that the last thing you do is put the eyepiece in when the telescope is upright. Start with the widest lens that will give you the widest field of view (32mm is a good choice) - that way when starting off you can see more stars, and you'll find it easier to figure out where you are using Stellarium's map.
- The Dobsonian is a reflector telescope, i.e. it has a big mirror at the bottom of the telescope tube (unlike the Northumberland and Thorowgood which are both refractors). Moisture may form on the mirror when you bring it inside and it's much warmer inside than outside - before replacing the cap on the telescope, make sure to wait for this moisture to evaporate away. You might have to come back in the morning if there's a lot of moisture.

First step to finding things in the night sky

Each of the paths given in this guide starts off by getting you to point the Dobsonian at a relatively easy target in the sky, usually a bright star in a constellation.

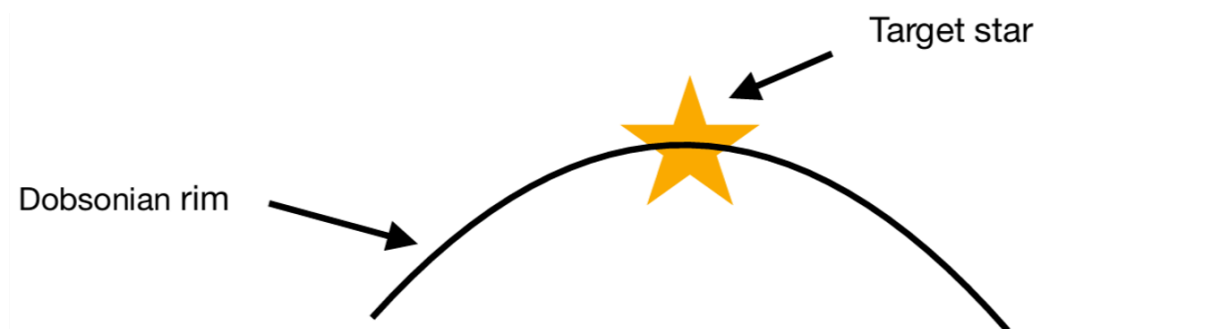


Fig. 1: What you should see when aiming at the target star (very definitely not to scale)

To do this, first move the telescope so that it's roughly pointing towards the target. Then, you'll have to crouch down near the base of the telescope tube, so that your nose is pretty much aligned with the upper rim of the telescope tube.

No, I'm not joking - this enables you to move the telescope, so that when you look across the telescope tube from this position, you should only just be able to see the target star.

Now you can have a look in the finder scope attached at the side of the Dobsonian to check whether you can see your target star. It might not be visible, so try moving the telescope around a little to see whether it enters your field of view.

If not, you'll have to crouch down again. If you do see your target star, then move the telescope carefully until the target star is aligned like so using the crosshairs in the finder scope.

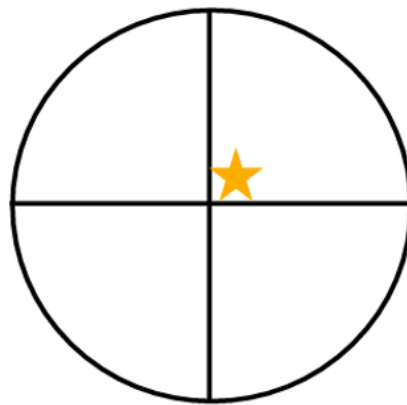


Fig. 2: What you should see in the finder scope when the star is roughly centred in the Dobsonian

Now when you look through the eyepiece, you should see the object roughly in the centre of your field of view. Success!

A note on what you'll see

The mirrors in the Dobsonian flip the image of the thing that you're trying to look at.

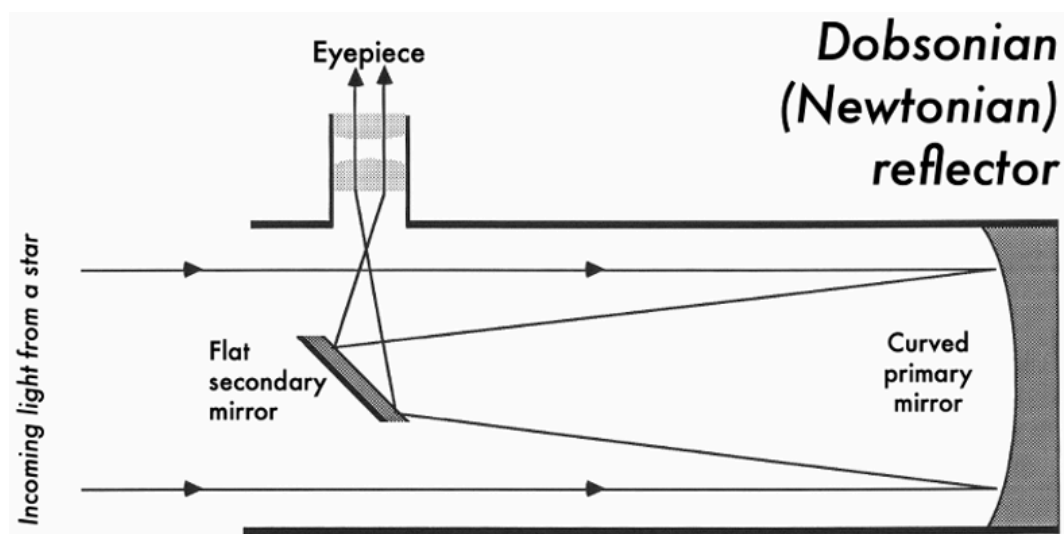


Fig. 3: The mirrors inside the Dobsonian

This has two main effects:

- Depending on what star maps you use, you might have to flip your star maps around so that what you're observing through the telescope matches what you see on the map. When using Stellarium, you'll have to lock your screen orientation before flipping your phone upside down.

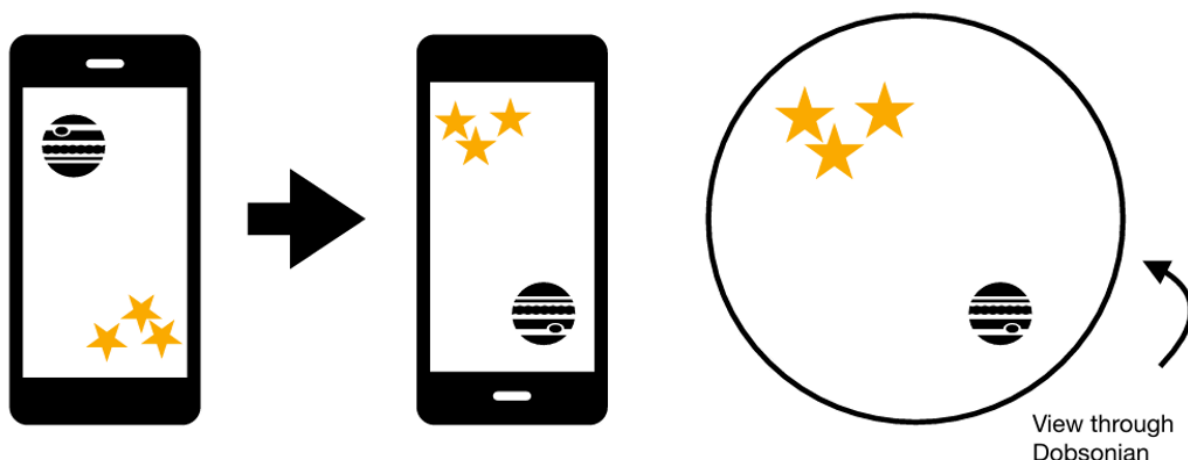


Fig. 4: You might have to flip your maps

A good way to check whether you've got your map orientated correctly is to use the constellation Cassiopeia, easily spotted as the 'big W'.



Fig. 5: Checking your map orientation using Cassiopeia (Stellarium)

Point the telescope at the star at the centre of the 'big W', Gamma Cassiopeiae, (yep you have to crouch, use the finder scope, etc.). Flip your phone around so the Stellarium map is upside down. Then, look at the surrounding stars (you may have to move the telescope around a little to check what stars are nearby). Look out for the triangle of stars circled above. You may have to tilt your phone slightly as well - just keep adjusting and moving until your map matches what you can actually see. To make sure you've got everything sorted, you should also check the position of the three bright stars close to Gamma Cassiopeiae. As you'll see, star hopping involves a lot of spotting shapes that surrounding stars make, not just your targets.

- Because of the flipped image, you have to move the telescope in the opposite direction in which you want your object to move when you look through the eyepiece/ finderscope. I thought about adding a more detailed description of what to do here, but honestly the only way to get used to it is to use the telescope - besides, you get used to it in no time at all.

A few final points

Star hopping is hard when you have a full/ gibbous moon - the brightness of both can drown out the fainter objects. The moon isn't included in this guide, but if you are planning on looking at it, make sure to use the lunar filter to avoid blinding yourself.

Now something that's obvious when you're told: The Earth rotates but the telescope doesn't. If you want to observe the same target for a while, then you'll have to keep adjusting the Dobsonian so that your object doesn't disappear from your field of view.

Also, getting used to Stellarium will take time, especially when using it as an actual star map for the first time. I remember having to get used to the difference between how big and bright stars appear to be relative to each other in Stellarium vs what I could actually see through the Dobsonian. All of this is just a matter of time and practice - you'll learn soon enough to recognise which shapes formed by stars that you can see are the same as the ones shown on your map, and if not you can always go back and start again – the stars will still be there :) (the paths in this guide use constellations that should be high in the sky all throughout Easter term).

Now's the fun part: the paths! Each has a rating:

★ = Easy

★ ★ = Medium

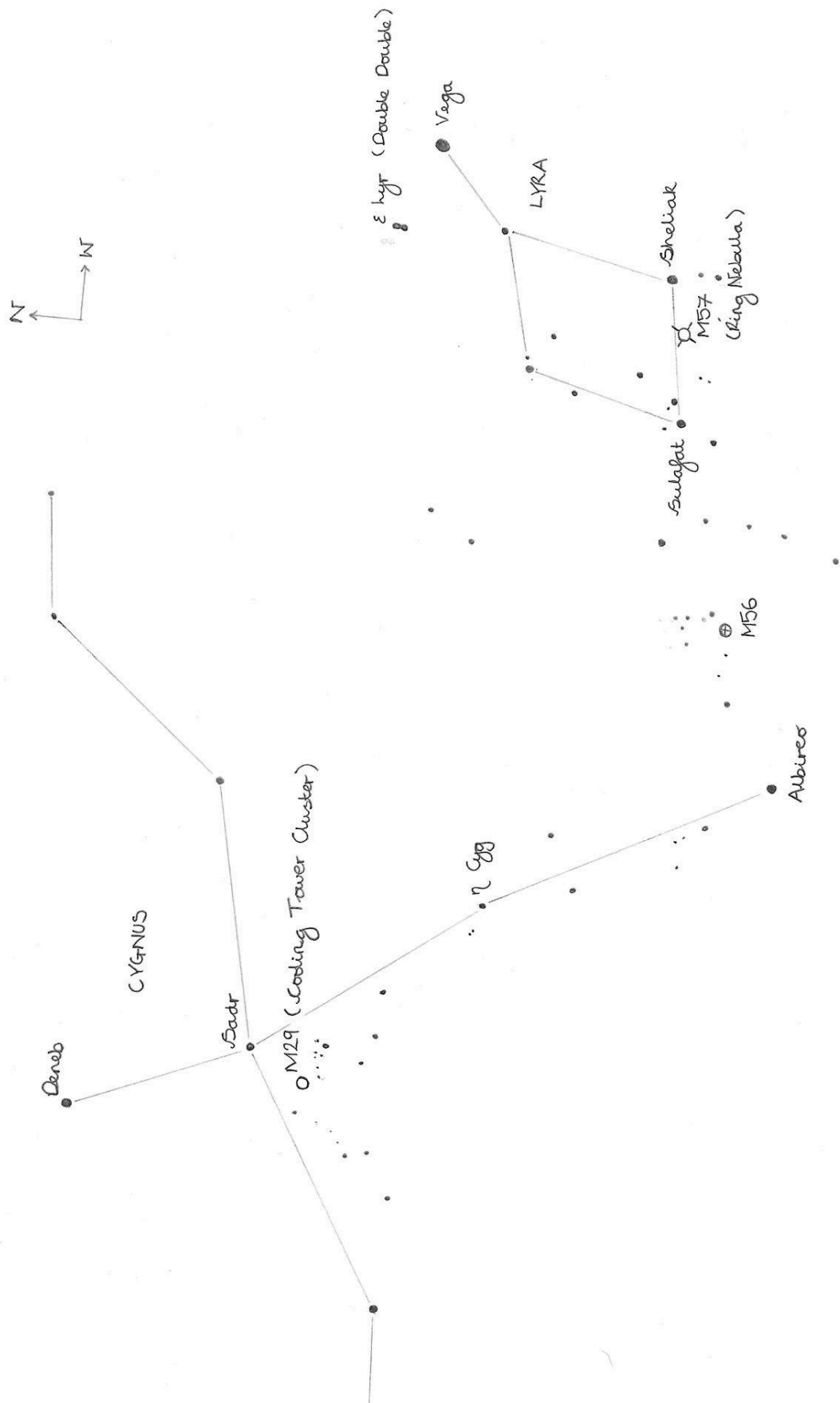
★ ★ ★ = Hard

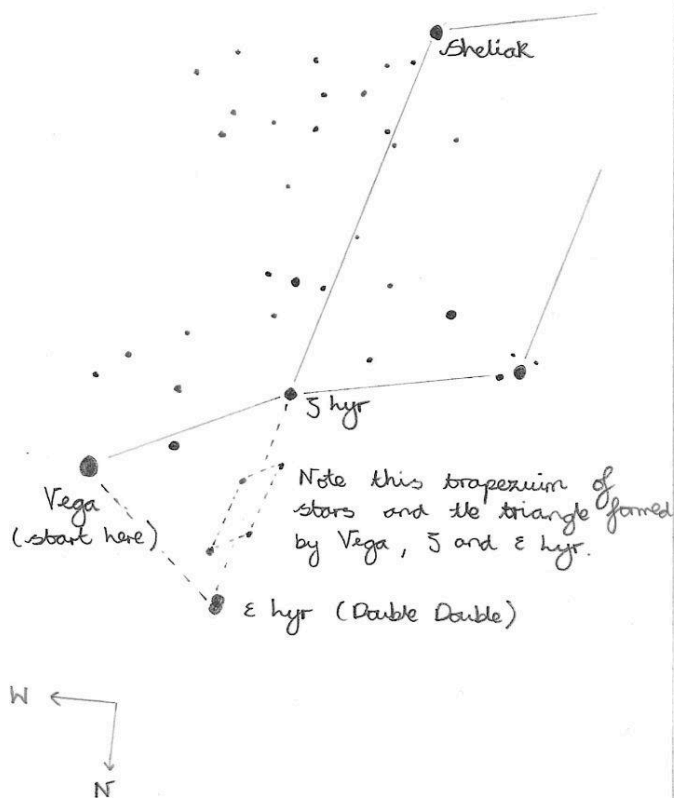
Have a go at the harder paths once you've got the hang of things, and see the *ObsSec Recommendations for Michaelmas 2026* when you get really good, as a lot of the fun comes from finding your own paths through the sky. For specific details to look out for within the objects you will see on the routes, and for some interesting facts about them, the *ObsSec Recommendations for Michaelmas 2026* guide will be of great use! (For Taurus and Auriga, see *ObsSec Recommendations for Lent 2026* guide.)

And with that, enjoy!

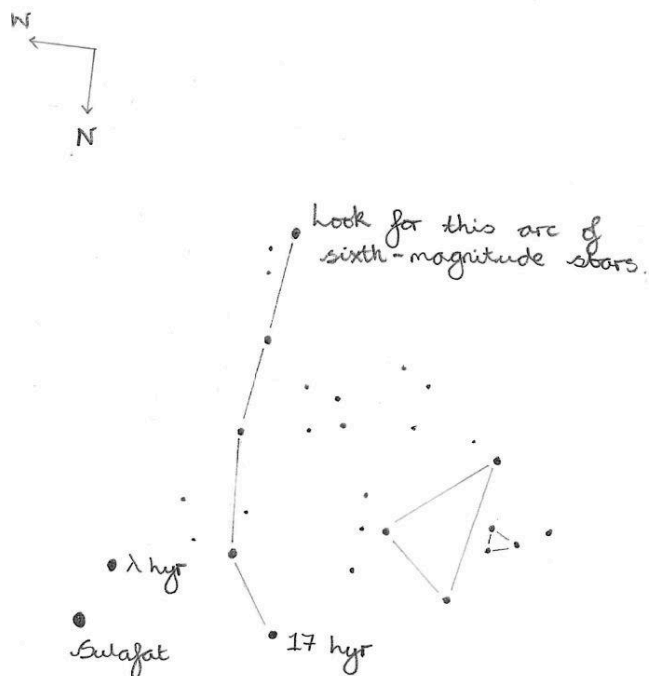
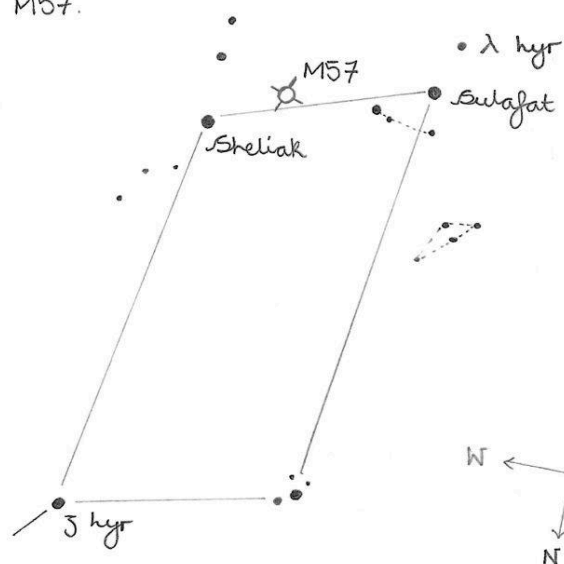
The Highlights of Cygnus and Lyra ★

Route by: William Moorhouse.

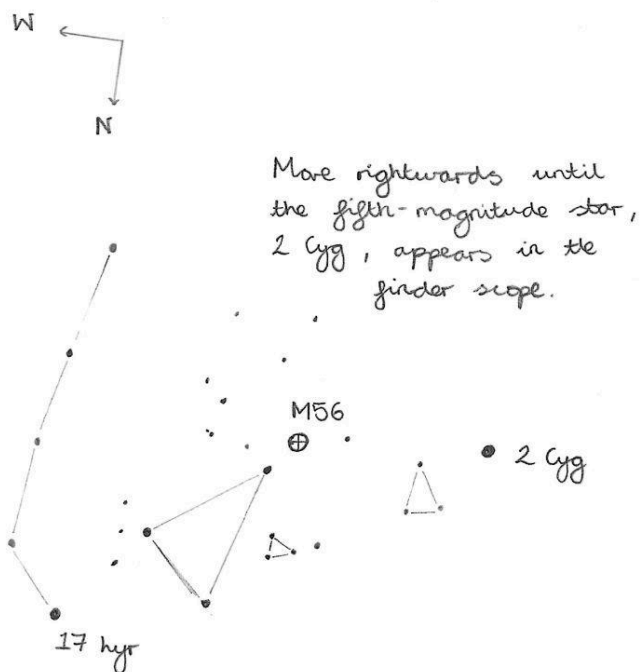




Centre Sheliak in the finder scope and move right, towards Sulafat, to find M57.

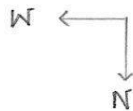


Place Sulafat in the bottom left corner of the finder scope field of view.



More rightwards until the fifth-magnitude star, 2 Cyg, appears in the finder scope.

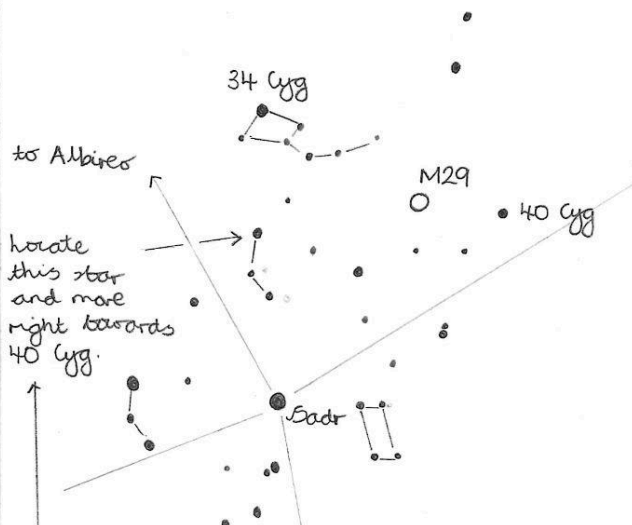
M56 is roughly halfway along a line from the two brightest stars in the field of view: 17 hr and 2 Cyg. Use the triangles of slightly dimmer stars to precisely locate it.



Continue moving right from M56 until the much brighter object, the double star system, Albireo, appears.

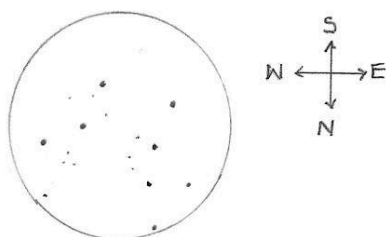
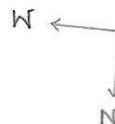
Yellow-orange primary
Albireo ● — Blue-white companion

Position the telescope such that Sadr is in the centre of the finder scope view.



locate this star and move right towards 40 Cyg.

This can be done using a low power eyepiece, as opposed to the finder. Both it and 40 Cyg are similarly bright (~ sixth-magnitude).

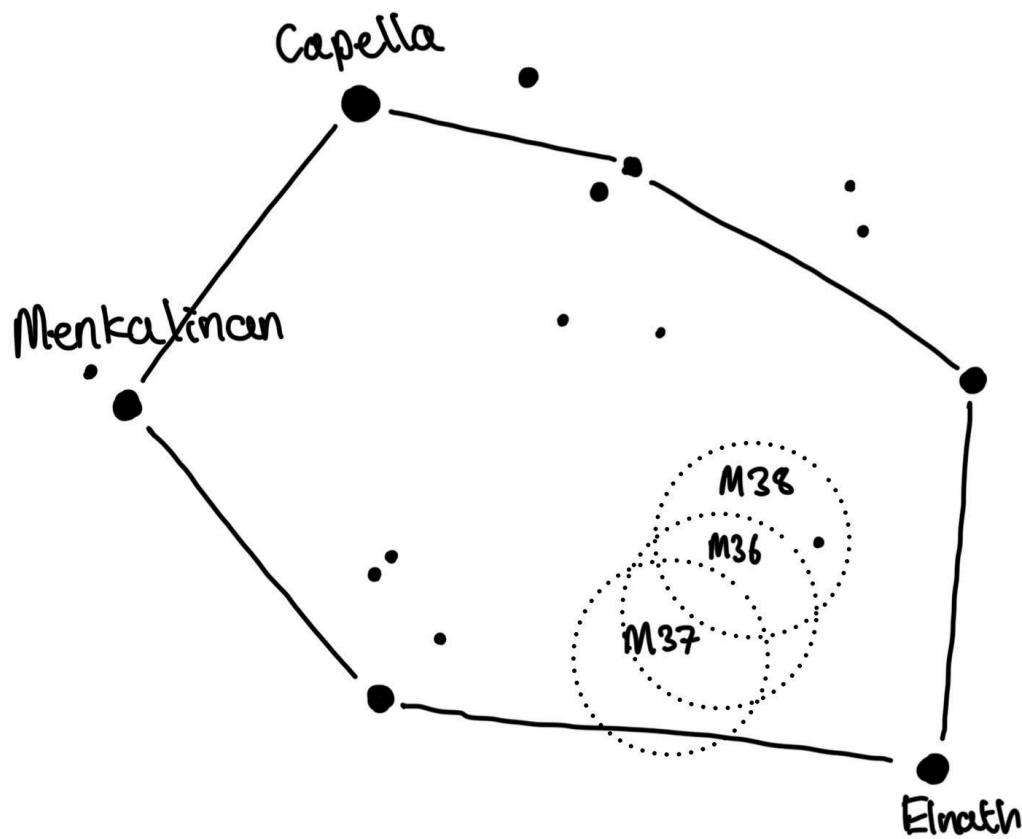


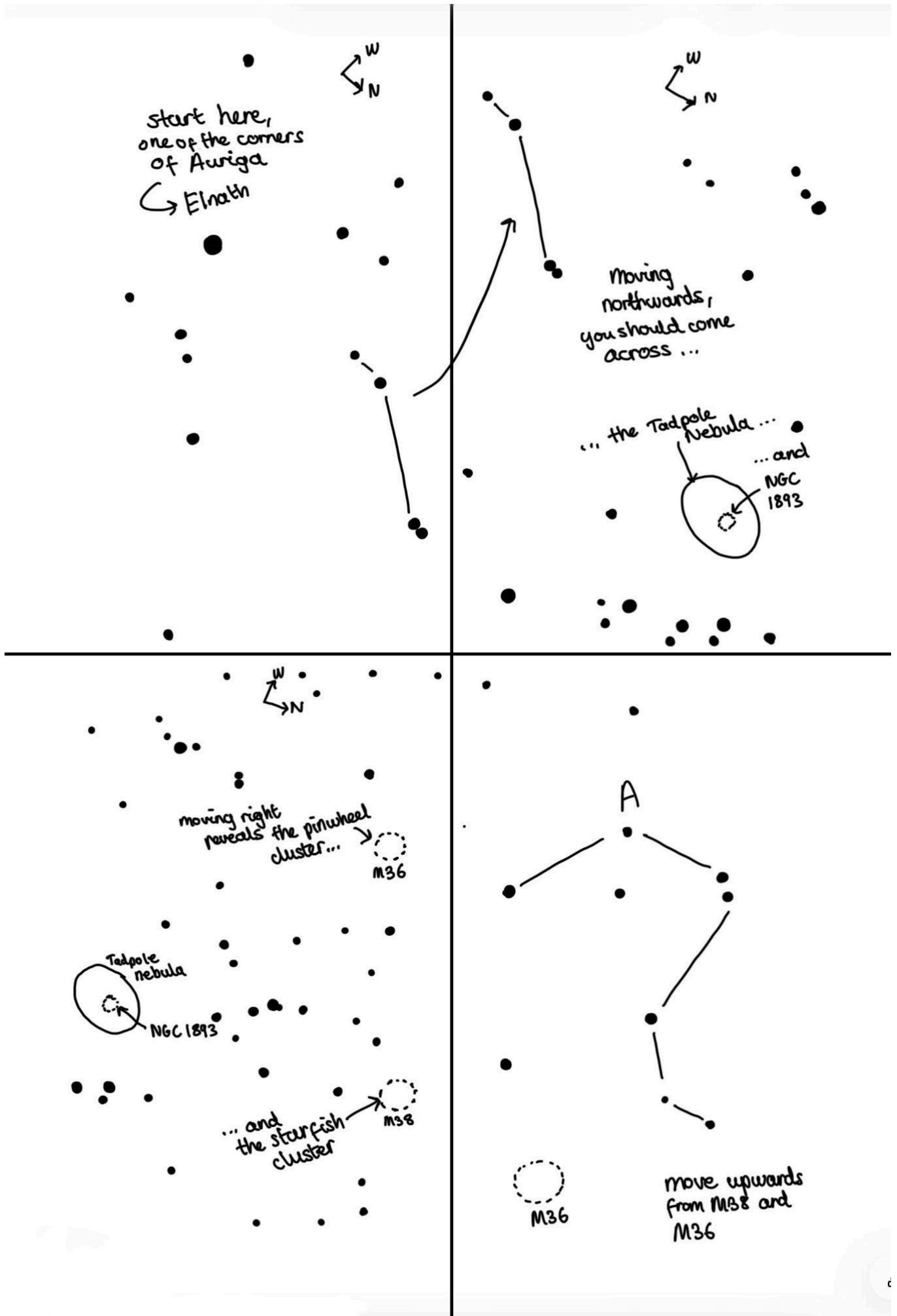
A sketch of what you can expect M29 to appear as in a high power field of view (i.e. 10mm eyepiece).

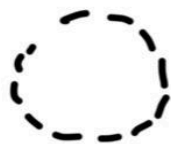
Open Clusters in Auriga ★

Route by: Lan Hall.

Auriga







M37

keep moving
upwards from
the hook to
see another
open cluster M37



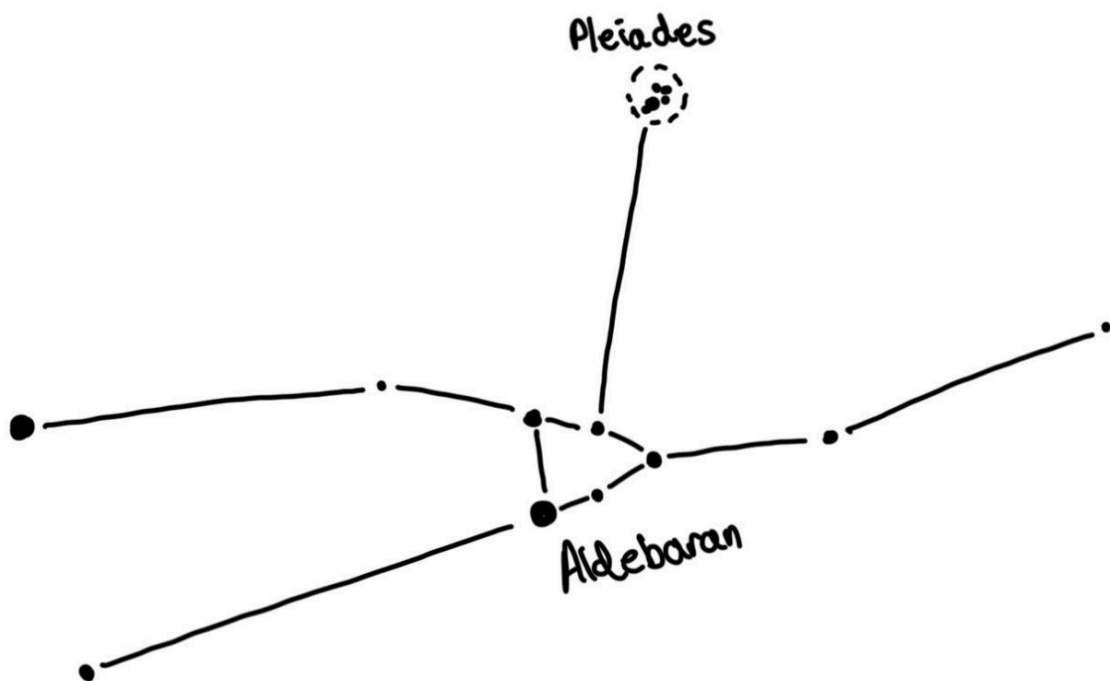
A

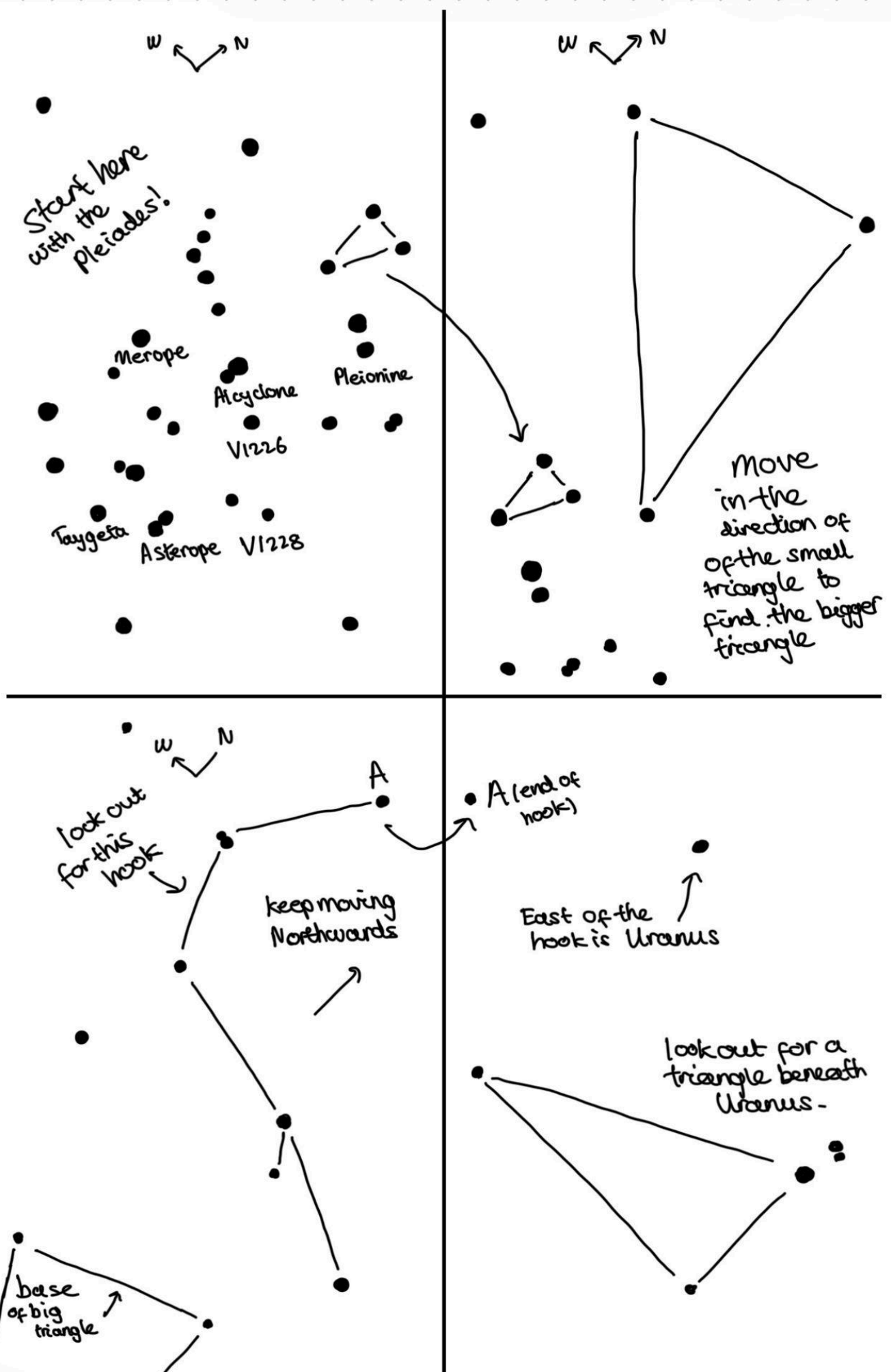


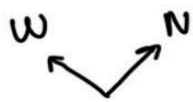
Taurus Highlights and Uranus ★★

Route by: Lan Hall.

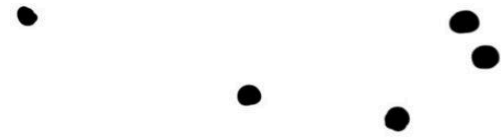
Taurus







V998



This time
start
here

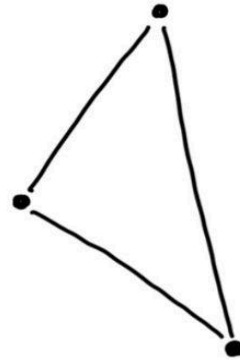
Aldebaran



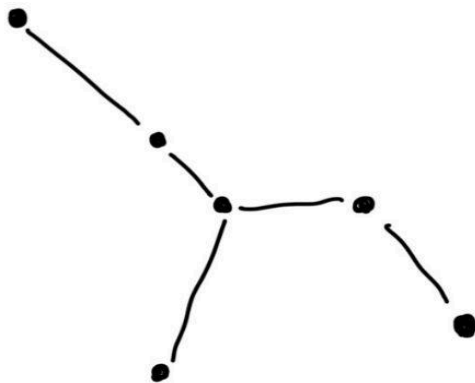
Aldebaran is the brightest
star in Taurus



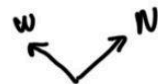
..



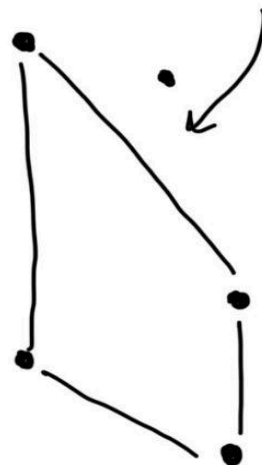
move
north east
to see this
triangle

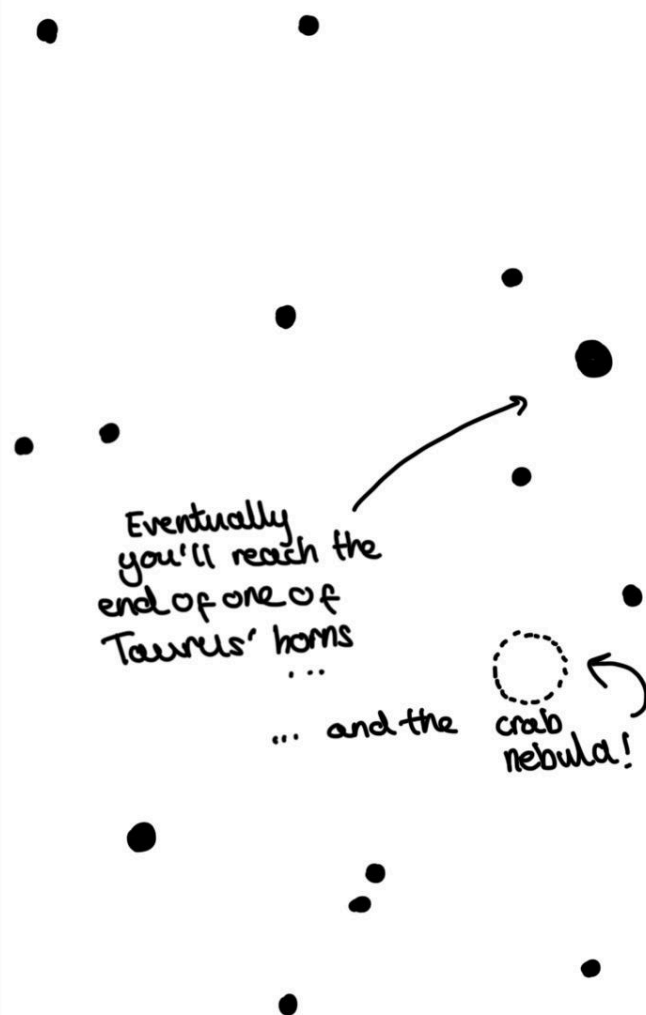
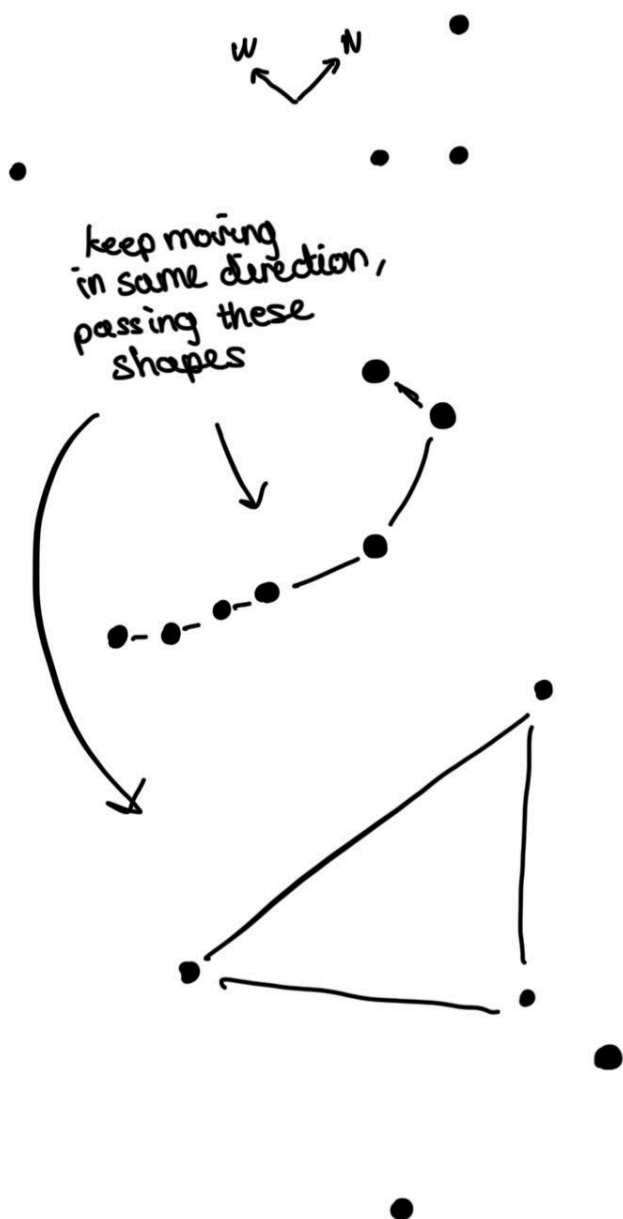


keep moving in the
same direction to
spot this shape



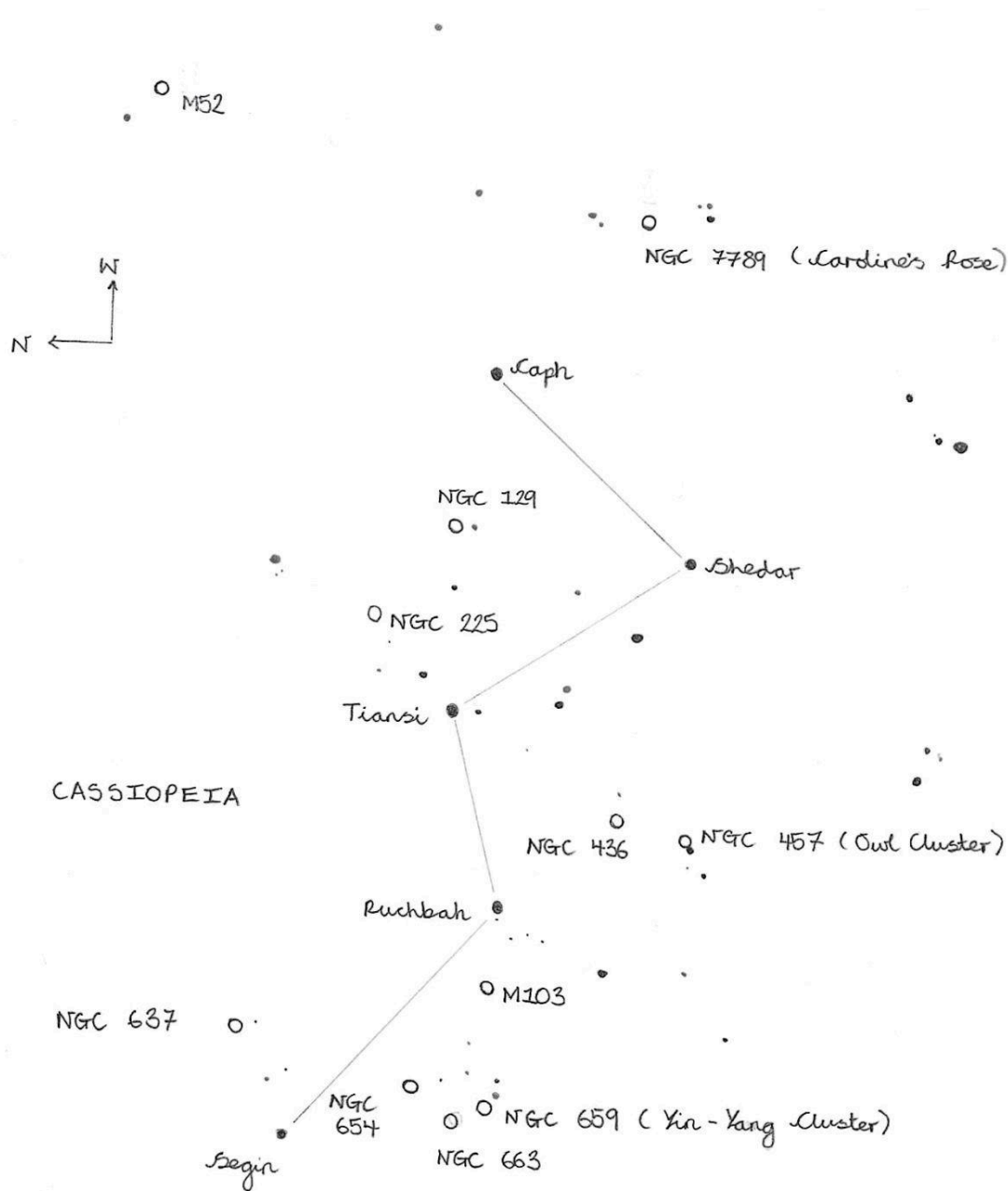
moving across → and
slightly more eastward
gets you to this
trapezium

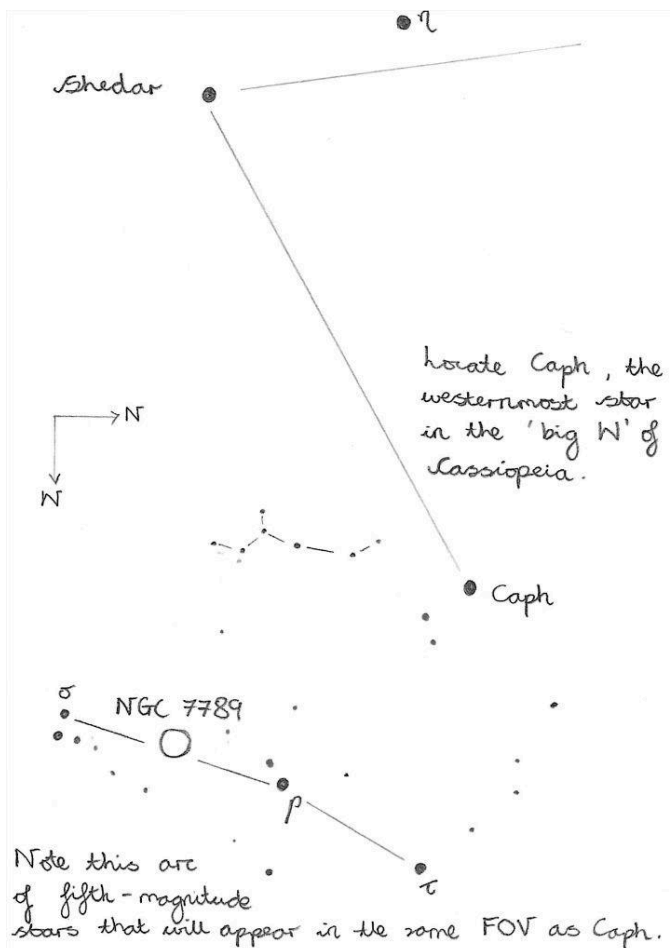




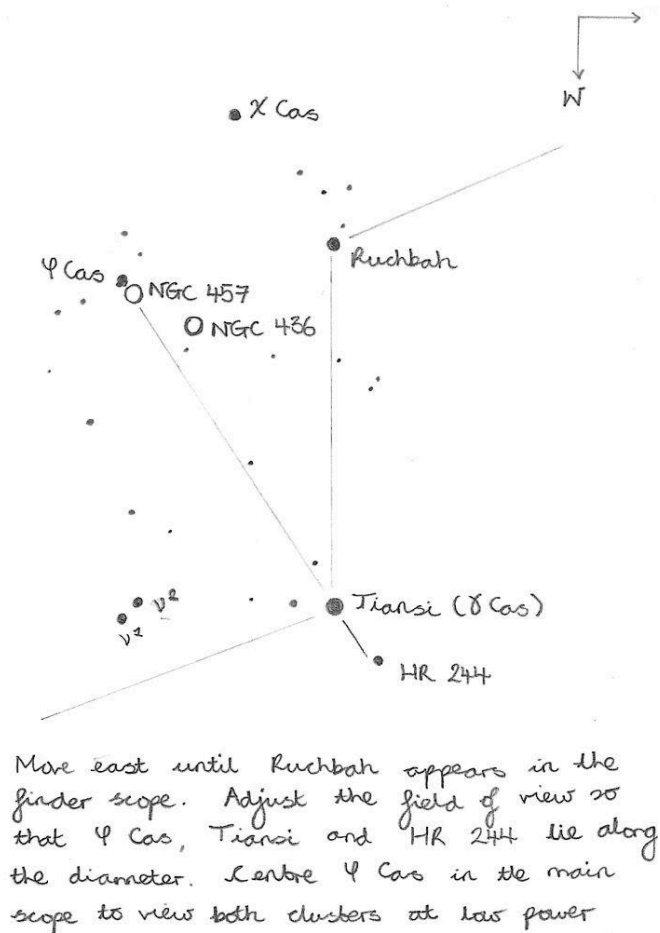
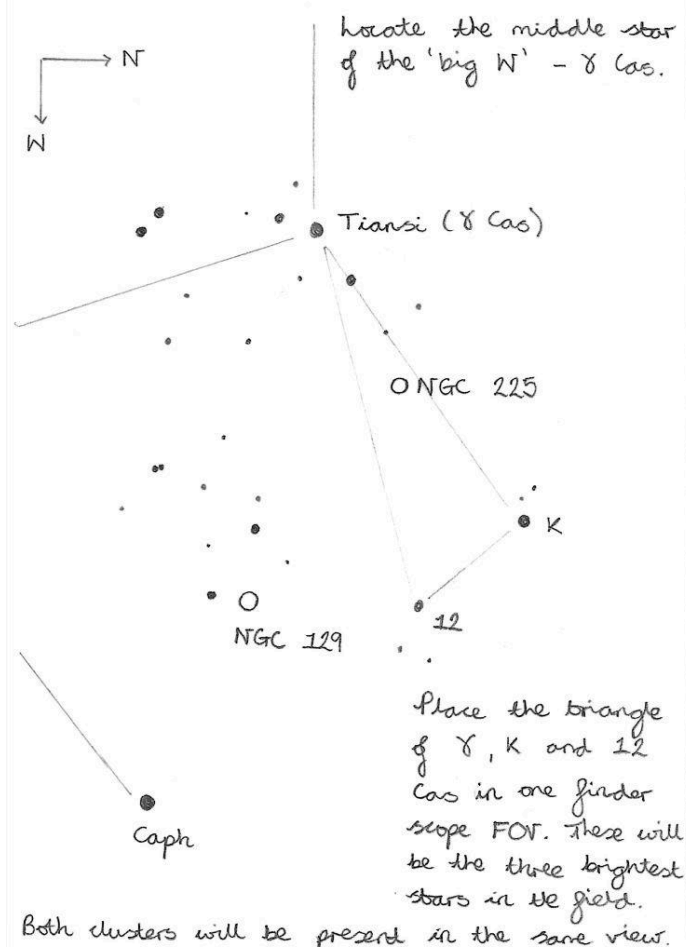
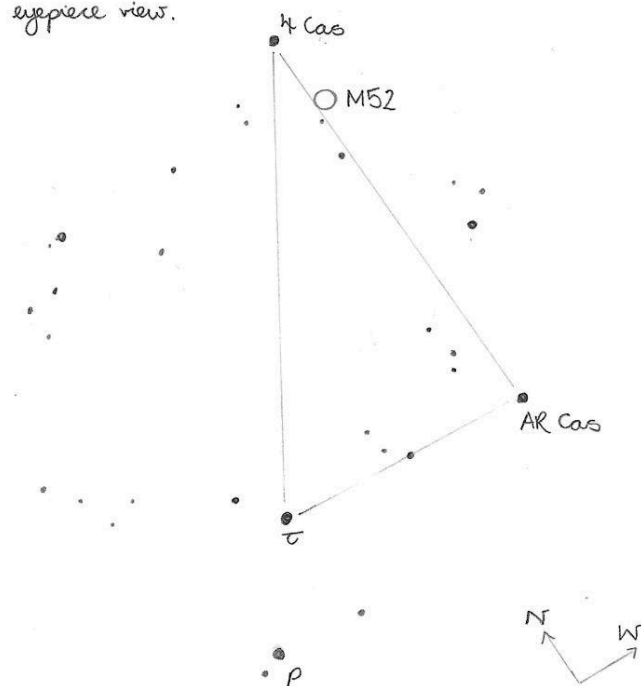
Open Clusters in Cassiopeia ★★

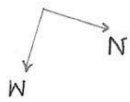
Route by: William Moorhouse.



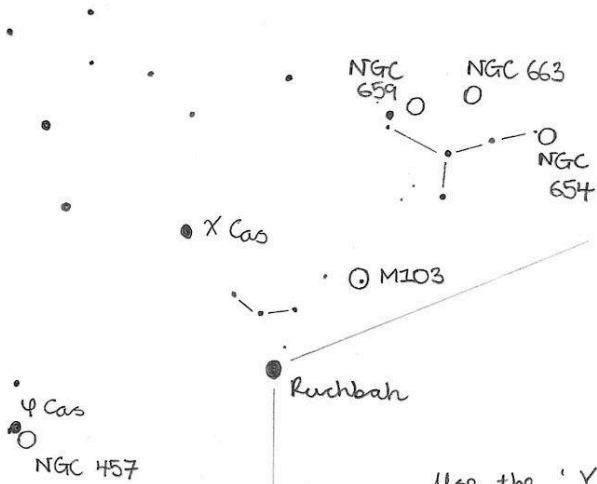
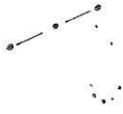


This right triangle of ϵ , τ and AR Cas will fill the finder scope field of view. Centre ϵ Cas - M52 will be visible within the same eyepiece view.



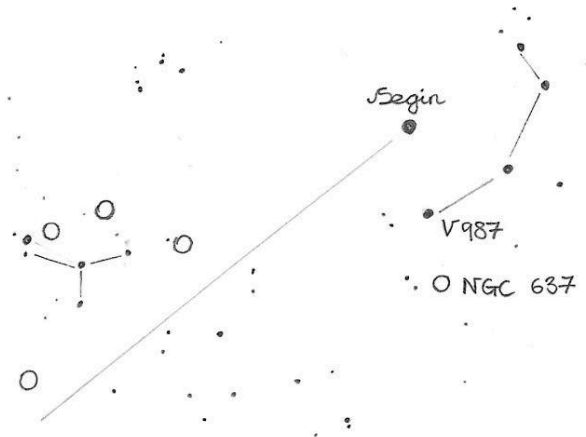
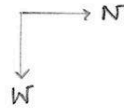


Position the scope so that Ruchbah and the line of three seventh-magnitude stars just fit into one field of view.



Use the 'Y' shape to place the three NGC clusters into a low power field of view.

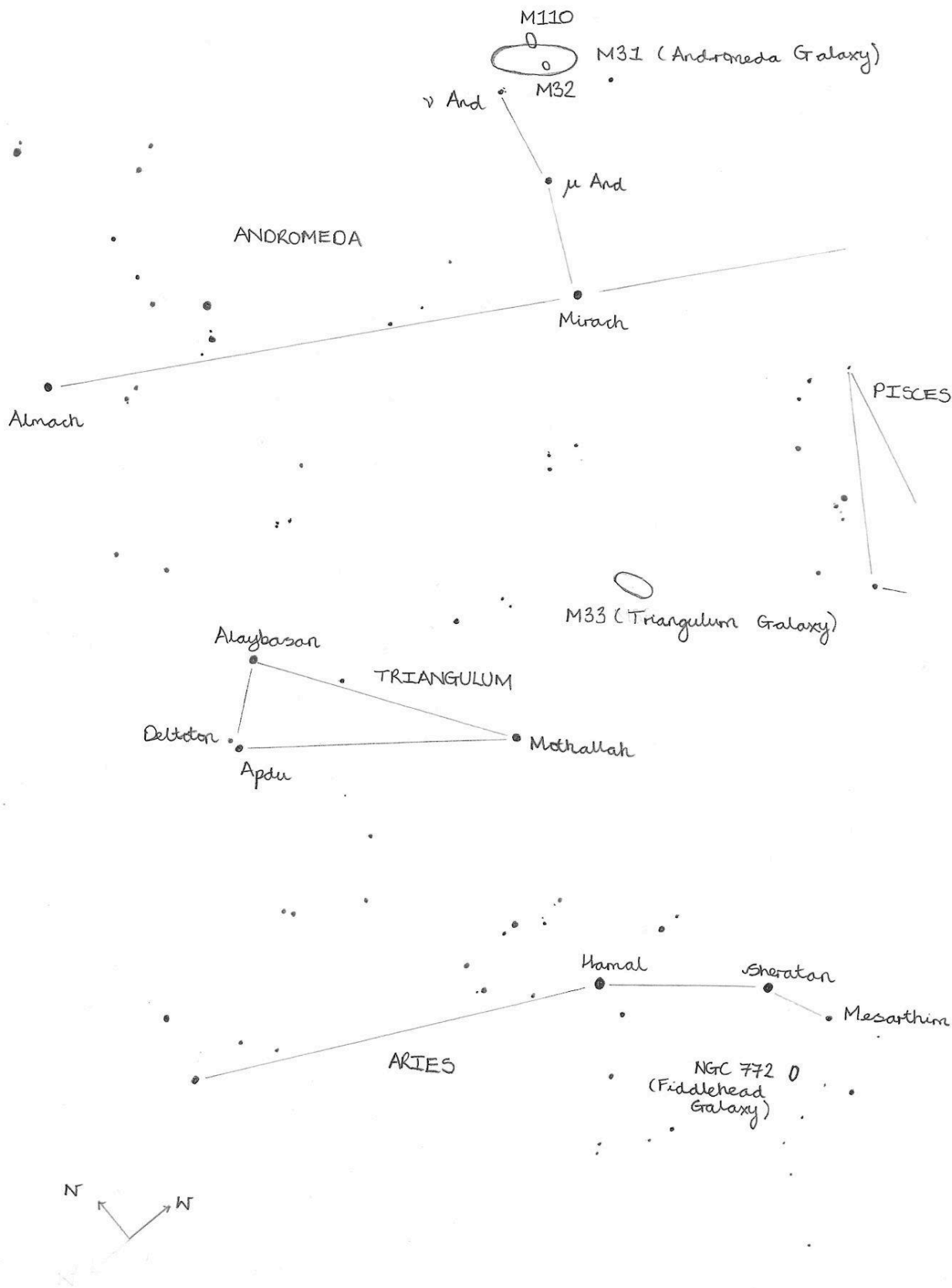
V987 and the two bright pairs of eighth-magnitude stars will fit in the same medium power FOV as NGC 637.

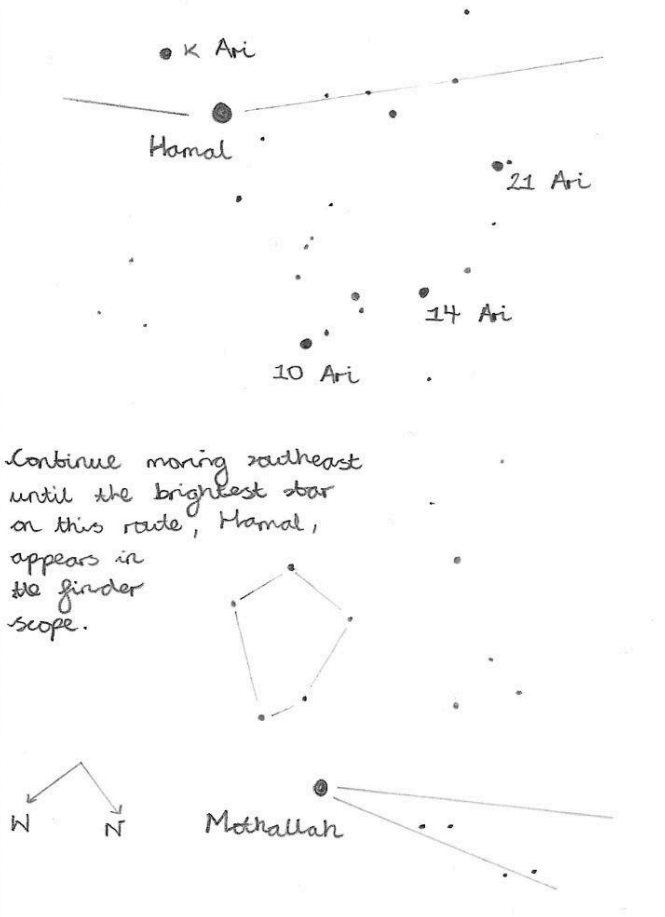
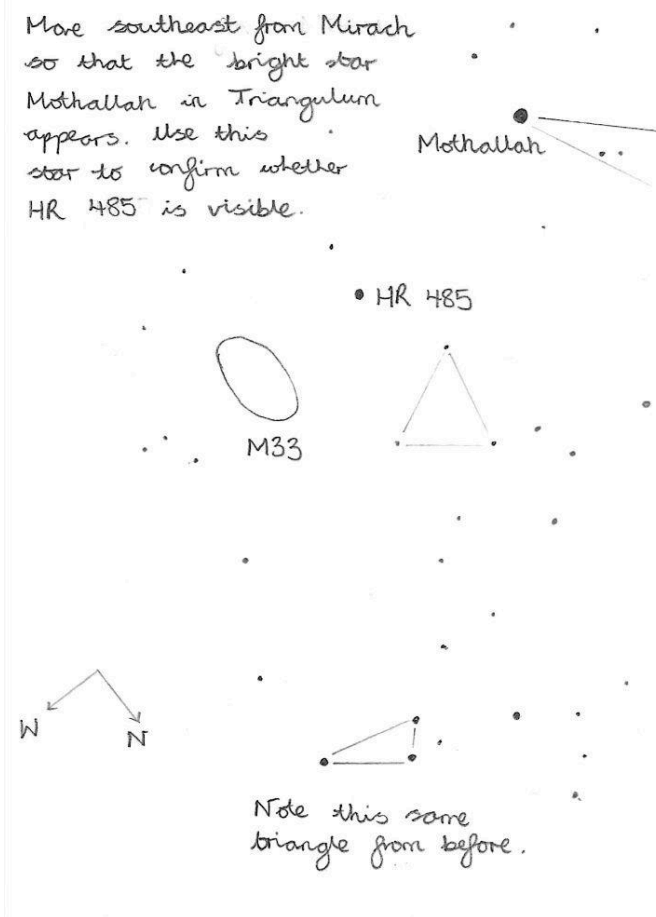
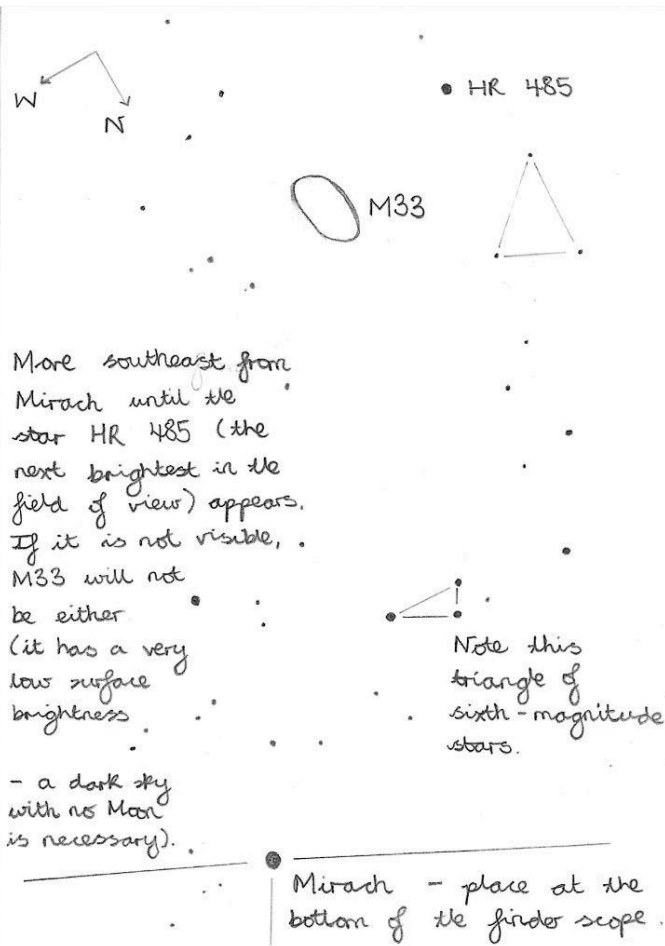
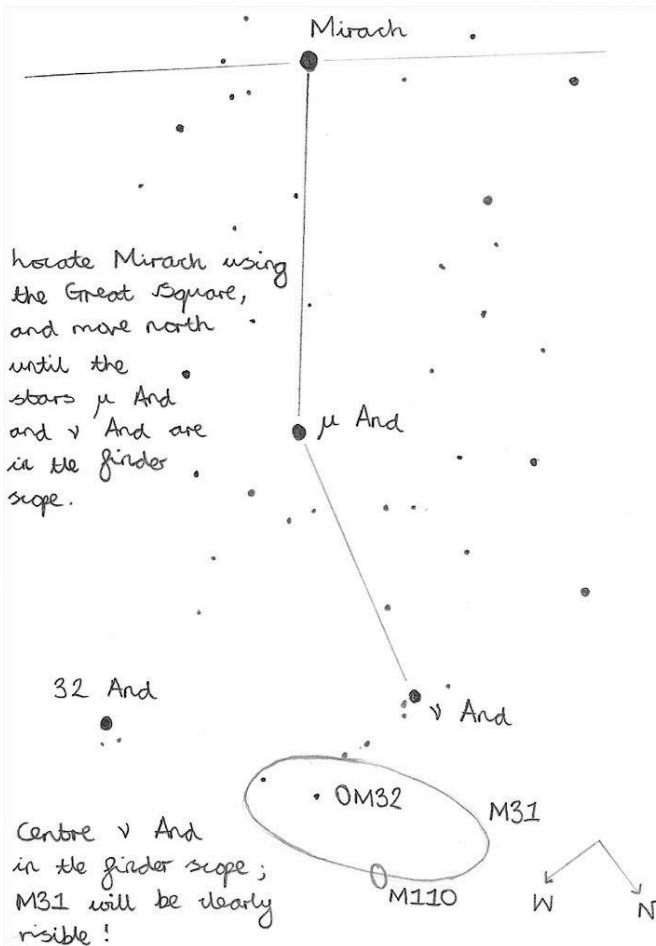


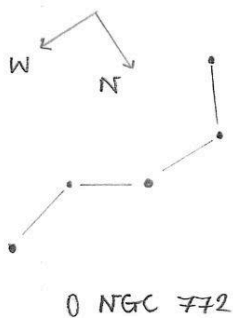
Place the 'Y' shape in the south of the finder scope field of view, so that Segin and its northern arc of four fifth-magnitude stars are present.

Local Group Galaxies and NGC 772 ★★★

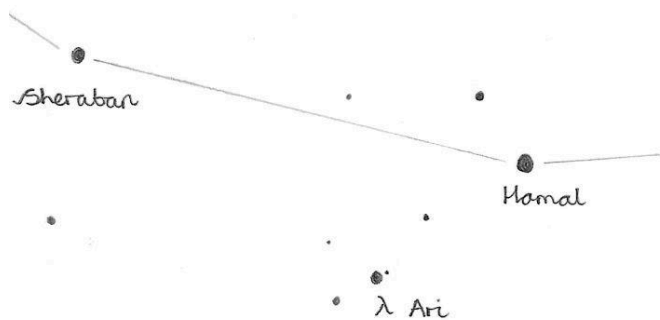
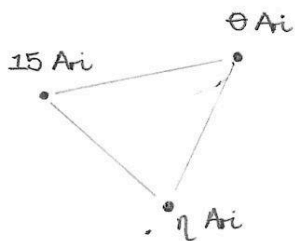
Route by: William Moorhouse.



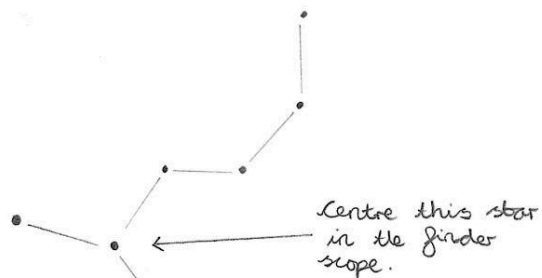
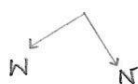




Note this triangle of
~ fifth-magnitude
stars.



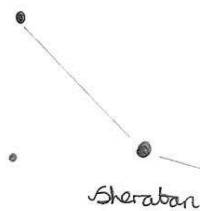
Place Hamal and Sheratan (the
brightest stars in Aries) in the same field
of view of the finder scope.



Centre this star
in the finder
scope.

NGC 772 - it is
a faint galaxy, which
may appear as a dim,
elongated disc even at
higher powers through the
main scope. Only try if
M33 was perfectly
visible.

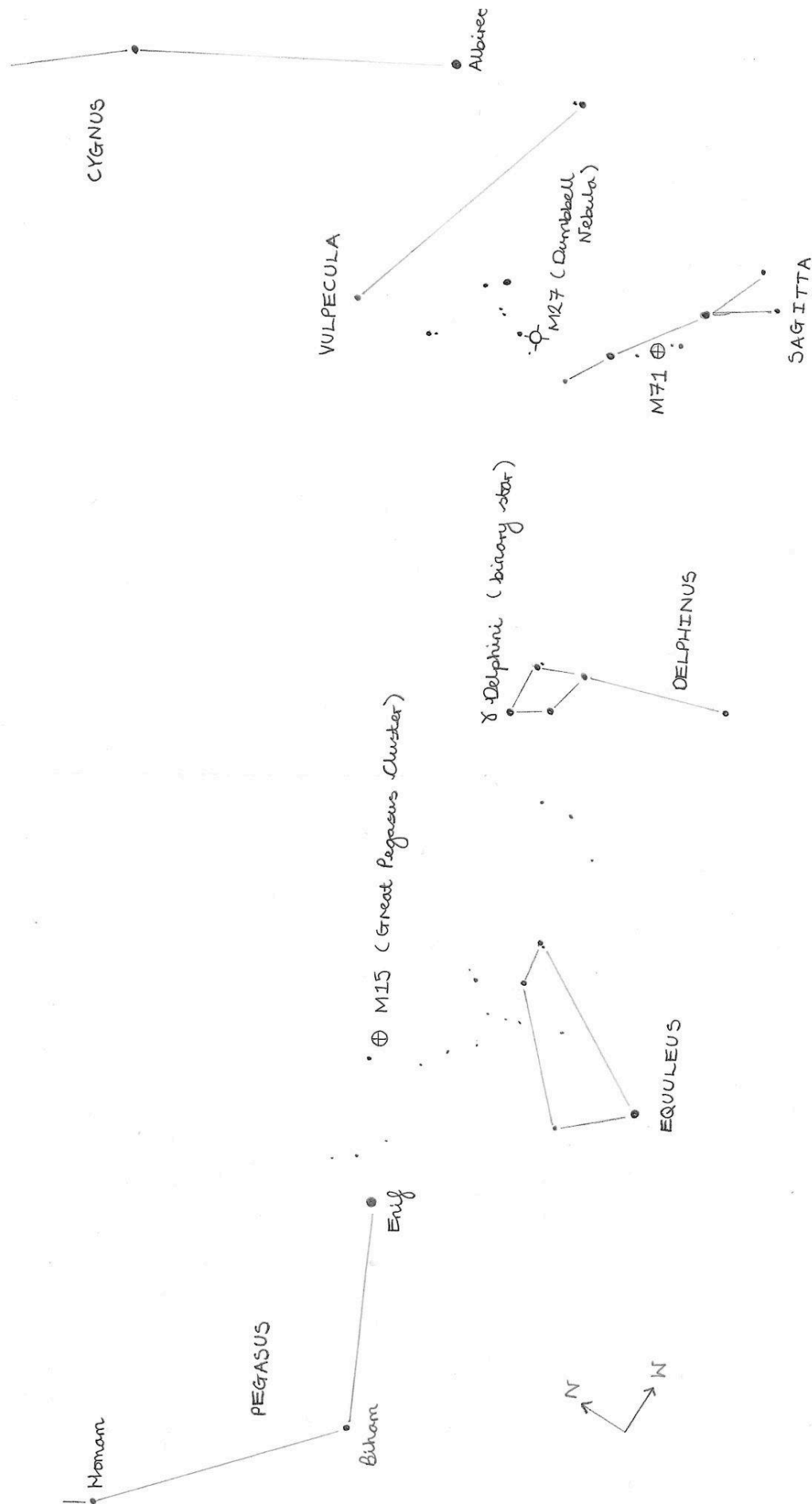
Mesarthim



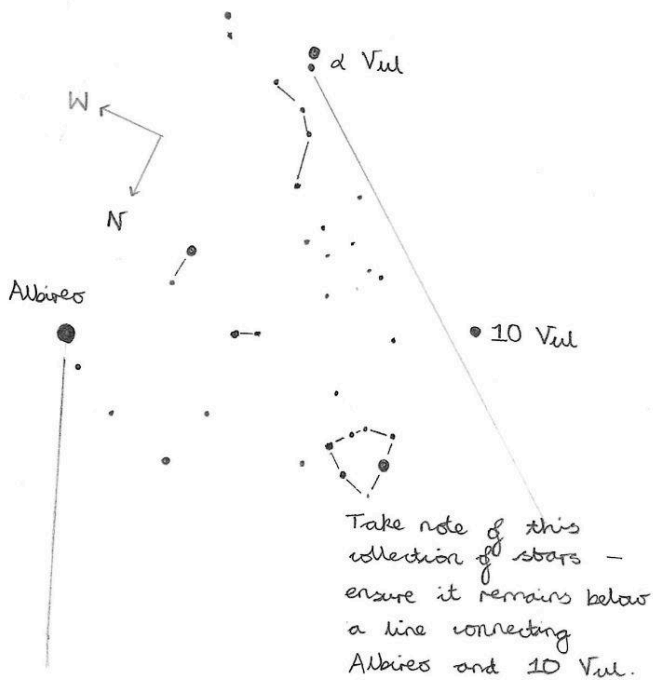
lambda Ari

From Cygnus to Pegasus ★★★

Route by: William Moorhouse.



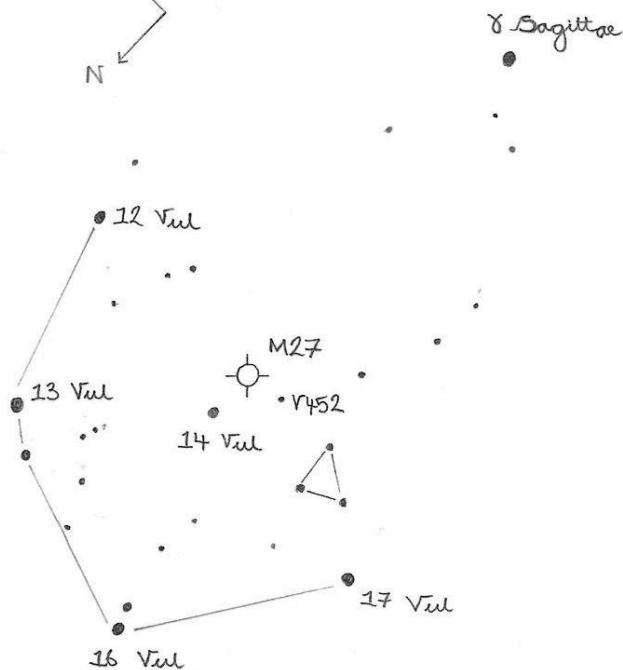
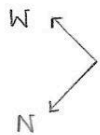
Start at Albireo, then move right until 10 Vul is in the far left of the finder scope view.



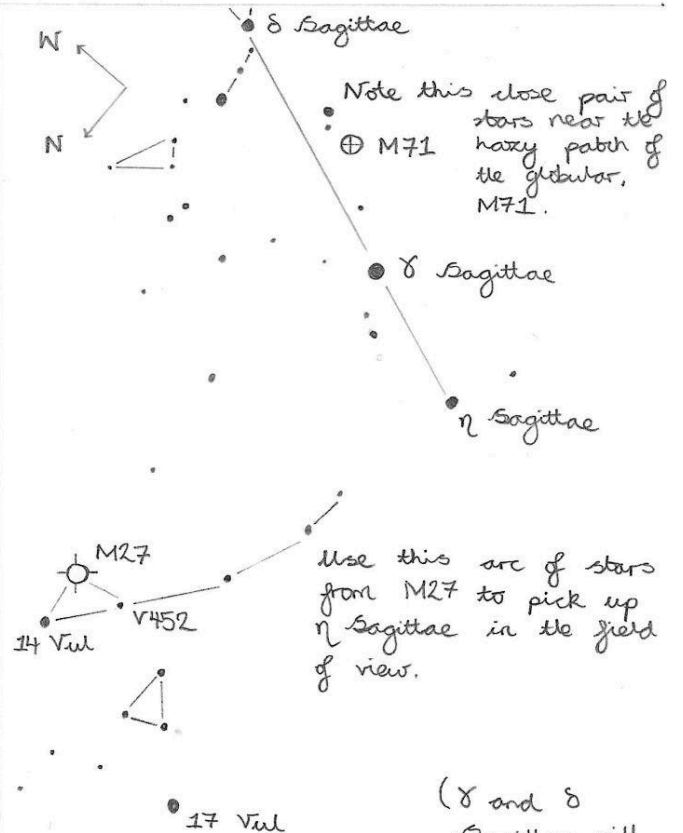
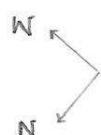
Albireo will be the brightest star in the triangle consisting also of α Vul and 10 Vul.



Continue moving right from 10 Vul.

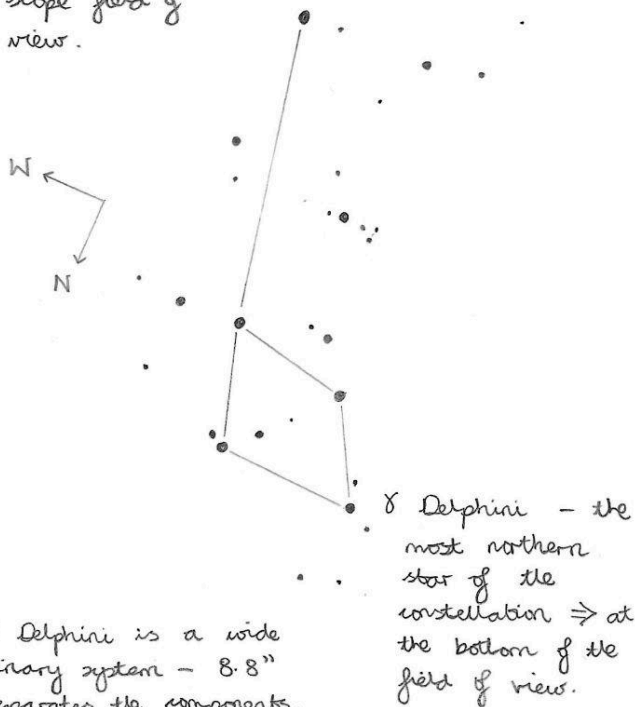


The triangle formed from the stars 14 Vul and V452 and from the Dumbbell Nebula (M27) will all fit within a medium power field of view (i.e. 15mm eyepiece).

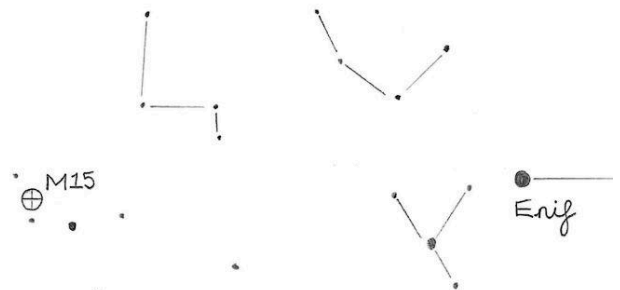


(δ and δ Sagittae will be the brightest stars in the finder scope.)

From M71, skew $\sim 30^\circ$ (six fields of view in the finder scope) to locate the constellation Delphinus. The five fourth-magnitude stars it contains will all fit in one finder scope field of view.



Position Enif (the brightest star on this route) in the right of the finder scope field of view. M15 will be on the left.



M15, a sixth-magnitude globular cluster, will be visible in the finder scope as a fuzzy spot, next to a star of a similar brightness.

Take note of these shapes in the sparse starfield between Enif and M15.

